

Variable time ($t_{Descharge} + t_{inspection}$) : (1.5 + 1) Volume dependent	This involves the visual and physical quality control of incoming merchandise.	Damage pallet verification.	($T_{Descharge} + T_{inspection}$) • X_i
Fixed time (T_F) Volume independent	Administrative inspection process.	Signing off on the warehouse management system.	25

Table 5. Classification of Total Time Components into Fixed and Variable Activities.

$$\text{Service time } (T_{si}) = T_F + (T_{Descharge} + T_{inspection}) \cdot X_i$$

Where X_i represents the number of pallets of the specific store, which directly modifies the unloading and inspection time at each node, for example (using a high-capacity vehicle):

- If this truck only delivers 2 pallets, the unloading and inspection time will only take 5 minutes $2 \times (1.5 + 1)$.
- If that same truck goes to another node and unloads 20 pallets instead, the worker will take 50 minutes $20 \times (1.5 + 1)$.

Another limitation for the maximum capacity in terms of euro pallets is defined by the following inequality:

$$\sum_{\text{Stores } i \in \text{Route}_k} X_i \leq Q_k$$

This constraint implies that the sum of all pallets from the stores visited by the truck k cannot exceed the maximum capacity Q_k of its vehicle type (*indicated in Table 2*).

IV. Mathematical Model

Decision variables

- Type of vehicle (low-, medium-, or high-capacity) according to the delivery route.
- The sequence in which the Lidl stores should be visited.
- Consequently, the number of vehicles and drivers required for the distribution process.

To support a more sustainable approach, additional decision variables are also considered, while keeping the main objective in focus.

- CO2 emissions generated by the selection of a vehicle category.
- Number of drivers (In accordance with recovery times).

Sets and Indices

Set	Index	Description
N	i, j	Total system nodes. Including all the existing nodes in the area (1 distributor, Dr. Oetker, 24 Lidl stores): $N = \{0, 1, 2, 3 \dots n\}, \text{ where } n = 25$
S	i, j	This subset isolates the stores from the central depot (node 0) within the global node set N . This subset is essential because the depot operates under entirely different logical constraints than the commercial destinations. While stores exhibit specific material demands ($X_i > 0$). This prevents the solver from generating redundant loops and increases efficiency by eliminating unnecessary “if” conditionals. $S = N \setminus \{0\} \text{ or } S = \{1, 2, 3 \dots n\}, \text{ where } n = 24$ This implies that the set excludes the origin node (Node 0).
K	k	A set of available vehicles inside the heterogeneous fleet, where $K = \{1, 2 \dots K\}$
M	m	Type (Category) of vehicle $M = \{"Low", "Medium", "High"\}$
K_m	$m \in M$	A disjoint subset of individual vehicles belonging strictly to the capacity category mentioned before. $\bigcup_{m \in M} K_m = K \quad K_m \cap K_{m'} = 0, \forall m \neq m'$ $\bigcup_{m \in M}$: Each and every truck in the fleet (K) belongs to one of the three capacity categories. There's no truck in the system considered left out of parameter restrictions $\cap = 0$: An individual truck cannot belong to two categories at once. It is physically impossible for a vehicle to belong to multiple categories simultaneously.

Parameters

d_{ij} : Distance in kilometers (km) between node i and node j . It is assumed that d_{ii} or $d_{jj} = 0$.

t_{ij} : Time of travelling in minutes (min) between node i and node j . Calculated based on the average speed of the road network in the area.

α : (Buffer-Traffic) Road safety and congestion scaling coefficient in the OWL region ($\alpha = X_i$:

Store demand. 1.15)

X_i : Discrete supply demand for each Lidl store $i \in S$ measured in quantity of pallets (units).

For the central depot evidently, is $X_0: 0$

v_{im} : Accessibility matrix (binary):

- 1 = If the vehicle m has physical access to the store $i \in S$
- 0 = If the vehicle is prohibited from accessing to that store i .

Q_m : Maximum pallet capacity ($Q_{Low} = 15$, $Q_{Medium} = 24$, $Q_{High} = 33$) for type of vehicle.

$S_{max m}$: The maximum number of stores that the type of vehicle m can visit in a single route.

u_{ik} : It is a continuous variable representing the number of pallets remaining in truck k .

immediately after leaving store i .

- 1 = If the vehicle m has physical access to the store i .
- 0 = If the vehicle is prohibited from accessing to that store i

For example, $v_{im} = 0$ would be when a high-capacity truck cannot have access to the historical center.

$D_{max m}$: Maximum route duration, depending on the driver's shift and type of vehicle m

expressed in minutes ($D_{Low} = 360$, $D_{Medium} = 480$, $D_{High} = 540$)

c_m : Variable gas cost in euros (€/km) depending on type of truck and emission penalty measured in euros per kilometer traveled. $c_{Low} = 0.25$, $c_{Medium} = 0.38$, $c_{High} = 0.56$

T_f : Fixed time of service for the administrative procedure. $T_f = 25$ in minutes.

$t_{discharge}$: Variable time in minutes for pallet discharge. (1.5 min/pallet).

$t_{inspection}$: Variable time in minutes for pallet inspection. (1 min/pallet).

Variable decision parameters:

- $y_{ijk} \in \{0, 1\}$ Binary variable for the route.
 - 1 = If the vehicle k goes directly from node i to node j .
 - 0 = Otherwise
- u_{ik} : Continuous variable of the subroute. Indicates the order of visit sequence of the store i by the truck k . $u_{ik} \geq 0$ to prevent isolated loops.
- $w_k \in \{0, 1\}$ Binary fleet usage variable.
 - 1 = If the vehicle k is activated to work.
 - 0 = If it is stored in the main deposit.

Objective Function

The goal of this mathematical function is to minimize the variable financial cost of distribution, which directly weights the environmental impact (CO2 emissions) resulting from the fuel consumption of the heterogeneous fleet over the distances traveled in direct proportion.

$$\text{Min } Z = \sum_{i \in N} \sum_{j \in N, j \neq i} \sum_{m \in M} \sum_{k \in K_m} c_m \cdot d_{ij} \cdot y_{ijk}$$

Breakdown:

1. $\text{Min } Z$: Optimization criterion is the minimization, which represents the economic and environmental cost of daily operation.
2. $\sum_{i \in N} \sum_{j \in N, j \neq i}$: Double summation over the network's adjacency matrix that checks every possible trip from origin node i to a destination node j . The condition $j \neq i$ ensures that a node cannot be treated as its own destination, preventing optimization from creating self-loops or assigning travel costs from a node to itself. Within the summation, the multiplication by the adjacency matrix acts as a filter: only origin–destination pairs connected by an existing network link contribute to the objective function, while non-connected pairs contribute zero. The double summation then aggregates the contribution of all valid origin–destination pairs across the network.
3. $\sum_{m \in M} \sum_{k \in K_m}$: Double summation for the heterogeneous fleet. It checks that each category of vehicle m (Low, Medium, High), and sequentially, each individual physical vehicle k belongs to the subset K_m .
4. c_m : Weighted Cost Coefficient, which not only represents the financial cost of fuel per kilometer but also proportionally penalizes the CO2 emissions factor.
5. d_{ij} : Physical distances in kilometers between nodes in the OWL region.
6. y_{ijk} : A routing variable that acts as the “switch” in the equation. If the truck k of type m travels within i and j , $y_{ijk} = 1$, the cost of that segment is added to the total ($c_m \cdot d_{ij} \cdot 1$). If the truck does not travel that distance, $y_{ijk} = 0$, nullifying the product and preventing any cost from being added to the system.

System Constraints

1. $\sum_{j \in S} y_{0jk} = w_k \quad \forall m \in M, \forall k \in K_m$
2. $\sum_{i \in S} y_{0ik} = w_k \quad \forall m \in M, \forall k \in K_m$
3. $\sum_{i \in N, i \neq 1} \sum_{k \in K} y_{ijk} = 1 \quad \forall j \in S$

4. $\sum_{j \in N, j \neq 1} y_{ijk} - \sum_{h \in N, h \neq j} y_{ijk} = 0 \quad \forall j \in S, \forall k \in K$
5. $\sum_{j \in N, j \neq 1} y_{ijk} \leq v_{jm} \quad \forall j \in S, \forall m \in M, \forall k \in K_m$
6. $\sum_{j \in S} \sum_{i \in N, i \neq j} X_j \cdot y_{ijk} \leq Q_m \cdot w_k \quad \forall m \in M, \forall k \in K_m$
7. $\sum_{j \in S} \sum_{i \in N, i \neq j} y_{ijk} \leq S_m^{max} \cdot w_k \quad \forall m \in M, \forall k \in K_m$
8. $\sum_{i \in N} \sum_{j \in N, j \neq i} (\alpha \cdot t_{ij}) \cdot y_{ijk} + \sum_{j \in S} (T_f + (t_{discharge} + t_{inspection}) \cdot X_j) \cdot (\sum_{i \in N, i \neq j} y_{ijk}) \leq D_m^{max} \cdot w_k$
 $\forall m \in M, \forall k \in K_m$
9. $u_{ik} - u_{jk} + Q_m \cdot y_{ijk} \leq Q_m - X_j \quad \forall i \in N, \forall j \in N, \forall j \in S, i \neq j, \forall m \in M, \forall k$
10. $0 \leq u_{ik} \leq Q_m \cdot w_k \quad \forall i \in N, \forall m \in M, \forall k \in K_m$

Variables:

$y_{ijk} \in \{0, 1\} \quad \forall i, j \in N, \forall k \in K$ It constrains the algorithm's search space to prevent absurd solutions from the solver, such as using "0.5 vehicles or -1 trips" between supermarkets.

$w_k \in \{0, 1\} \quad \forall k \in K$ Therefore, routing and fleet usage decisions are binary, whereas the sequential sub-route counters are positive continuous variables.

$u_{ik} \geq 0 \quad \forall i \in N_0, \forall k \in K$

Indices

$i \in N$: Index representing the origin node of a route (which may be the distribution center (DC) or a Lidl store).

$j \in N$: Index representing the intermediate destination node or arrival stop visited by a vehicle.

$h \in N$: Secondary index representing the subsequent destination node (departure node) to which a vehicle travels immediately after completing a delivery.

Explanation of System Constraints

R1: Mandatory exit from the Central Depository

Origin node is fixed in $i = 0$ (Dr. Oetker Central Depository). The summation evaluates all possible destinations of the set of store, and the same equation is repeated for each vehicle k inside the heterogeneous fleet K_m .

- If vehicle k is activated ($w_k=1$), consequently $\sum_{j \in S} y_{0jk} = 1$.
- If the vehicle is not used ($w_k = 0$), the restriction means that the truck cannot enter or exit from the deposit.

R2. Mandatory Return to the Central Depository

The destination node is fixed in $j = 0$ (Dr. Oetker Central Depository). The summation evaluates all possible origin nodes of the set of stores from which the vehicle can return.

- The truck is required to close its cycle by generating one arc leading back to the central depot. This ensures that no route remains open once deliveries are complete.

R3: Total Coverage and Single Customer Visit

By setting this constraint strictly equal to 1, the solution space is restricted so that each store is visited exactly once by a single truck. If an inequality (≥ 1) were used instead, the model would allow two different trucks to visit the same store to split the order, which would violate Lidl's logistics policy of avoiding duplicate loading dock costs and administrative paperwork.

R4: Flow Conservation at Commercial Nodes

It is evaluated for every combination of store $j \in S$ and vehicle $k \in K$. The index i represents the nodes entering j , while the index h represents the nodes departing from j .

- **Active Transit (1-1= 0)** : If vehicle k is assigned to serve store j from an origin node i , the incoming binary variable is activated ($y_{ijk} = 1$). To satisfy the equality to zero, the solver is then forced to activate exactly one outgoing binary variable ($y_{jhk} = 1$). This ensures route continuity by requiring the vehicle to proceed to a subsequent destination node h .
- **Inactive (0 - 0 = 0)**: If the optimal route of vehicle k does not include store j , all binary variables associated with that node take the value zero. The flow balance is therefore satisfied by default, allowing the vehicle to travel through other regions of the network without affecting the connectivity matrix of the unvisited nodes.

R5: Accessibility physical restriction

- If store j (for example, in an OWL-restricted historic city center) prohibits high-capacity trucks, the parameter is set to $v_{j^{High}} = 0$. The inequality then becomes ≤ 0 , which immediately eliminates any inbound arc to that store served by a heavy vehicle.
- If the store has permitted access $v_{jm} = 0$, the inequality then becomes ≤ 1 , making the solver freely decide if that route can be used or not.

R6: Fleet Maximum Load Capacity Constraint

This constraint prevents the physical overloading of a vehicle.

- If truck k visits store j ($y_{ijk} = 1$), the store's pallet demand is accounted for $X_j \cdot 1$. The summation, therefore, accumulates the total pallet demand of all stores assigned to the route of that specific truck.
- If the truck is inactive ($w_k = 0$), the right-hand side becomes 0, ensuring that no load is assigned to it. If the truck is active ($w_k = 1$), the upper bound is strictly set to the nominal capacity of its corresponding vehicle category $Q_m = (15, 24, 33 \text{ pallets})$.

R7: Commercial Stop Limit per Route Constraint

This constraint counts a value of "1" each time an arc leading to a commercial store is activated. The sum of all inbound commercial arcs, therefore, represents the exact number of delivery stops assigned to a route.

- By bounding this quantity with $S_m^{max} \cdot w_k$, the solver is prevented from generating excessively long routes that could lead to driver fatigue or compromise product freshness. The maximum allowable number of stops is adjusted according to the corresponding truck category m .

R8: Traffic-Adjusted Workday Time Window

- If truck k visits store j , the corresponding assignment variable equals 1, activating the service time at that location. The service time consists of a fixed administrative component plus a variable unloading and inspection time that depends on the number of pallets delivered to the store. This service time is added to the total driving time of the route.
- To account for traffic congestion in the OWL region, all travel times are multiplied by a safety factor of $\alpha=1.15$, increasing the nominal travel times by 15%. The resulting total route duration (including both driving and service times) must not exceed the maximum allowable working time for the driver. This upper bound is enforced only when the truck is active ($w_k=1$); otherwise, the right-hand side becomes zero, ensuring that an inactive vehicle cannot be assigned a route.

R9: Load Propagation Constraint

The continuous variable tracks the amount of cargo remaining on a truck as it moves along its route.

- If a connection between two locations is not part of the selected route, this constraint has no practical effect and does not influence the solution.

However, when the truck actually travels from one location to the next, the constraint ensures that the remaining load is updated correctly. After serving a store, the truck must carry fewer

pallets than before, with the reduction matching exactly the number of pallets delivered at that stop.

As a result, the truck's load decreases progressively along the route, guaranteeing that deliveries are made consistently and that the remaining cargo is always physically feasible. This prevents the model from creating unrealistic routes in which the truck's load does not change after completing deliveries.

R10: Residual Load Boundary Conditions

This constraint defines the valid range of the residual load carried by a truck at any point along its route. The remaining load can never be negative, since a truck cannot deliver more pallets than it carries. Likewise, it can never exceed the truck's maximum loading capacity.

- If a truck is not used, the constraint forces its residual load to remain zero throughout the model. Consequently, only active trucks are allowed to carry cargo, while inactive trucks are assigned no load.

Solver Approach

Solver and version: *

Model Type: Mixed-Integer Linear Programming (MILP)

The model is classified as mixed because it combines binary decision variables, which determine whether a truck travels between two locations, with continuous variables, which represent the amount of cargo remaining on each truck throughout its route (Fachrizal, 2020). The problem is computationally complex because it simultaneously involves deciding how to assign customer demands to vehicles while determining the optimal sequence of deliveries. These decisions combine characteristics of the *Bin Packing Problem* and the *Traveling Salesman Problem*.

IMPORTS

```
import json
import time
import os
import random
import copy
import math
```